CS & IT ENGINEERING



Theory of Computation

- Based on Problems
- Grammar
- PDA
- CFL
- Closure Properties

(15 - 20)

Context Free Languages

DPP 01 Discussion Notes



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[MCQ]

#Q. Consider the following grammar:

$S \rightarrow abSba \mid bSb \mid aSa \mid a \mid b \mid \in$

What is the language generated by above grammar?

A CFG that generates all palindromes over alphabet (a, b).

B CFG that generates all palindromes over alphabet (a, b) that do not contain substring “ab”.

C CFG that generates all palindromes over alphabet (a, b) that contain substring “aa”.

D CFG that generates all palindromes over alphabet (a, b) that do not contain substring “aa”.

Palindrome:

Order of accessing does not change while reading from beginning or end.

Ex $L = \{a, b, \epsilon, aa, aba, bab, abba, \ldots\}$

$S \to aSa \mid bSb \mid a \mid b \mid \epsilon$

a S a
a b S b a
a b a b a
a b b b a

{ odd length palindrome

[MCQ]

#Q. Consider the following grammars:

G1: S → aSc | $ B
B → bBc | $

G2: S → aaScc | $ B
B → bBc | $

G3: S → aSc | $ B
B → bbBcc | $

Which of the following is true regarding G1, G2 and G3?

A L(G1) ⊆ L(G2), L(G2) ⊆ L(G3)

B L(G2) ⊆ L(G3), L(G1) ⊆ L(G3)

C L(G1) ⊆ L(G3), L(G3) ⊆ L(G2)

D L(G2) ⊆ L(G1), L(G3) ⊆ L(G1)

$S \to aSb \mid \in$

$a^n b^n \mid n \geq 0$

Push Push Pop

skip

$L(G_1) = a^m \$ b^n \$ c^n c^m \Rightarrow a^m \$ b^n \$ c^{n+m} \mid m, n \geq 0$

$\Rightarrow$

$L(G_2) \subseteq L(G_1)$

$L(G_3) = B = b^{2n} \$ c^{2n}$

$\Rightarrow a^m \$ b^{2n} \$ c^{2n} c^m$

[MCQ]

#Q. Consider the following context free grammar:

$S \rightarrow aA \mid aBB$

$A \rightarrow aaA \mid \epsilon$

$B \rightarrow bB \mid bbC$

$C \rightarrow B$

What will be the equivalent simplified CFG for the given grammar?

A $S \rightarrow aA \mid a$
$A \rightarrow aaA \mid aa \mid b$

B $S \rightarrow aA \mid a$
$A \rightarrow aaA \mid aa$

C $S \rightarrow aAa \mid B$
$A \rightarrow aaA \mid aa$
$B \rightarrow bB \mid bb$

D $S \rightarrow aAa \mid B$
$A \rightarrow aA \mid b$
$B \rightarrow bB \mid bb \mid a$

(S, Ab, ab)

(1) $S \to aA \mid aBB \mid a$

$A \to aaA \mid aa$

$B \to bB \mid bbC$

$C \to B$

$\{ S \to aA \mid a$

$A \to aaA \mid aa \}$

(2) $S \to aA \mid aBB \mid a$

$A \to aaA \mid aa$

$B \to bB \mid bbB$

bB, bbB, bbB, bB

[MSQ]

#Q. Consider the following grammar:

S → AB

A → BaB | a

B → bbA

Which of the following is true regarding given grammar?

A. Every string of the above grammar have at least two a's.

B. Every string have three consecutive a's.

C. Every string have alternate a and b.

D. Every string have b's in multiple of 2.

[MCQ]

#Q. Given the following two grammars:

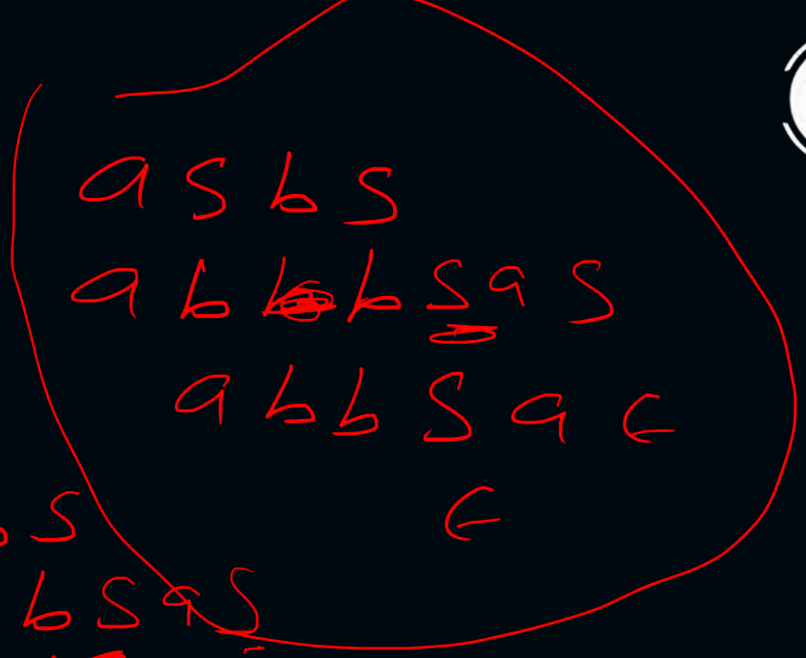
G1: $S \rightarrow AB \mid aaB$

$A \rightarrow a \mid Aa$

$B \rightarrow b$

G2: $S \rightarrow aSbS \mid bSaS \mid \in$

What is true regarding above grammars?

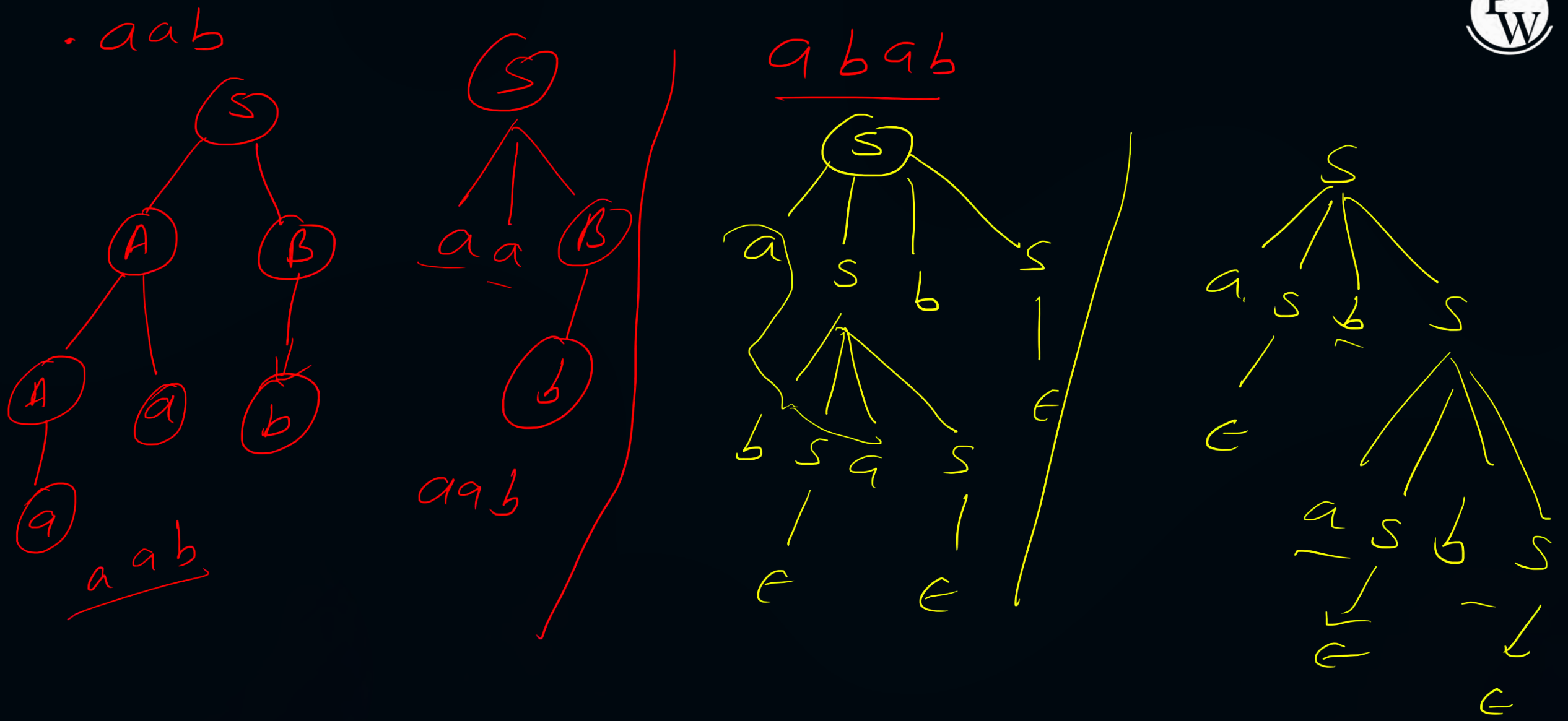


A. Both G1 and G2 are ambiguous.

B. Both G1 and G2 are unambiguous.

C. Only G1 is ambiguous.

D. Only G1 is unambiguous.


aab
S
A
B
A
a
b
a
aab
S
a
a
B
b
aab
abab
S
a
S
b
S
b
S
a
S
ε
ε
ε
S
a
S
b
S
ε
a
S
b
S
ε
ε

[MCQ]

#Q. Consider the following languages

$L1 = \{a^n b^{n+m} c^m \mid n, m \geq 0\}$

$L2 = \{a^p b^q c^{p+q} \mid p, q \geq 0\}$

Which of the following is true?

Non-Regular, DCFL, CFL, CSL

Non Regular DCFL

Push Push

A. Only L1 is Regular. ✗

B. Only L2 is Regular. ✗

C. Both L1 and L2 are Regular. ✗

D. None of these ✓

[MSQ]

#Q. Which of the following is string accepting mechanism of PDA.

A. PDA using final state. ✓

B. PDA using empty stack. ✓

C. PDA using both empty stack and final state. ✓

D. PDA using transition state. ✗

a, b, c

[MSQ]

#Q. Which of the following is correct push operation:

A $\Sigma(q, a, b) = (q', ab)$

B $\delta(q, a, b) = (q, ab)$

C $\delta(q, a, b) = (q', ab)$

D $\Sigma(q, a, b) = (q', \in)$

[NAT]

#Q. Consider the following statements:

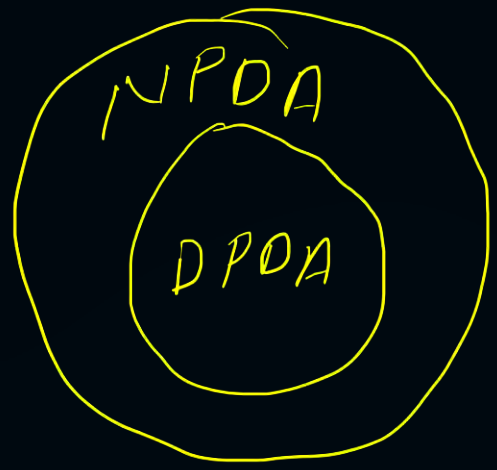
(I) All DPDA are NPDA.

(II) All NPDA are DPDA.

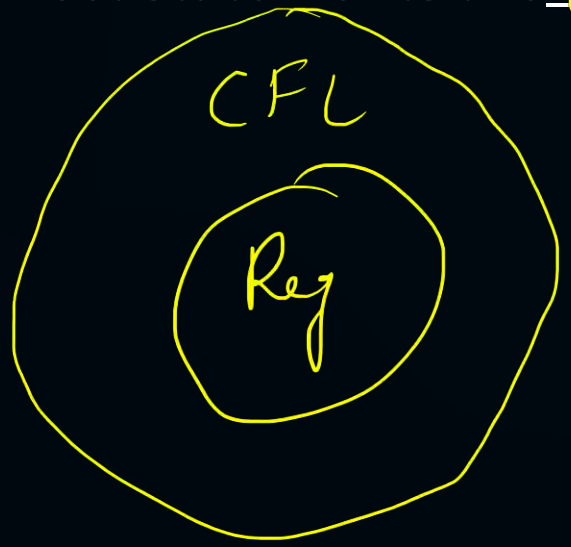
(III) All NPDA and DPDA are equivalent.

(IV) All context free language are regular language.

The total number of correct statements are ____.



DCFL → CFL



[MCQ]

#Q. What does following transition means:

$\delta(q, \in, b) = (q', b)$



A Push b

B Pop b

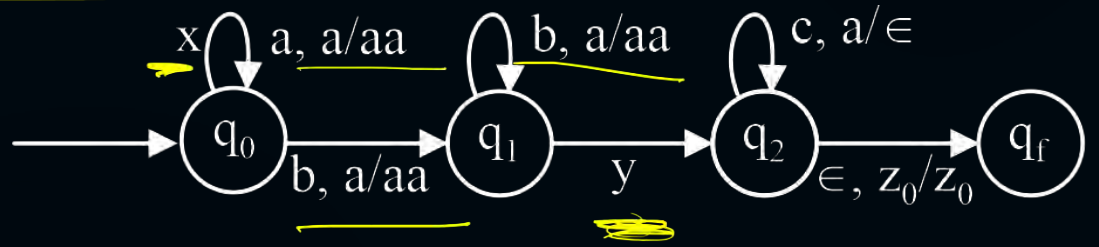
C Read b

D No operation

[MCQ]

#Q. What are the values of x and y, if the language accepted by NPDA is

$L = \{a^m b^n c^{(m+n)} \mid m, n \geq 1\}$.



A $x = \in, a/a$; $y = c, b/\in$

B $x = a, z_0/z_0$; $y = c, b/c$

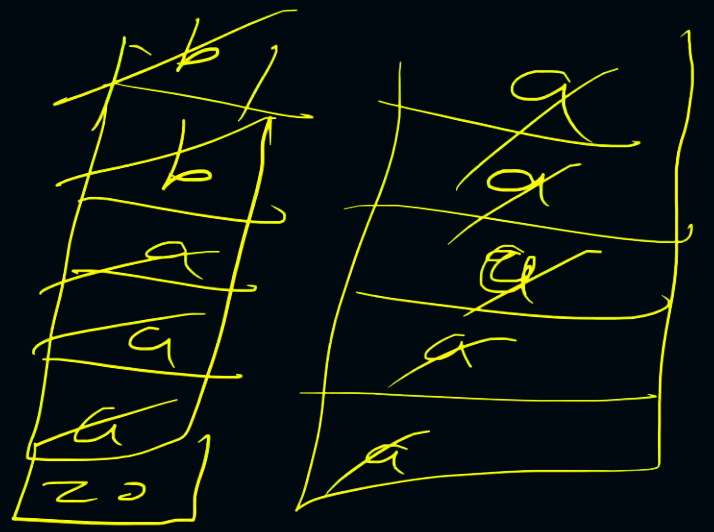
C $x = a, a/aa$; $y = b, c/c$

D $x = a, z_0/az_0$; $y = c, a/\in$

$a^m b^n c^{m+n}$

$a^m b^n c^n c^m$

$a^3 b^2 c^2 a^3$



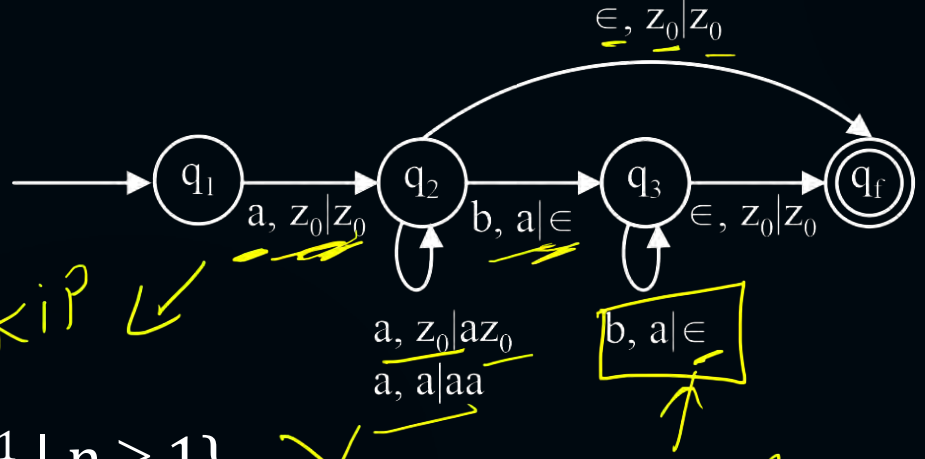
q ↳ b, a | ba

Push

a, Z_0 | a Z_0

[MCQ]

#Q. Which language is accepted by the following PDA.



(A) $L = \{a^n b^{n+1} \mid n \geq 1\}$

(B) $L = \{a^{n+1} b^m \mid n, m \geq 1\}$

(C) $L = \{a^{n+1} b^n \mid n \geq 0\}$

(D) $L = \{a^n b^{n+1} \mid n \geq 0\}$

[MSQ]

A, D

#Q. Which of the following languages are accepted by PDA.

CFL

push pop skip

$a^n b^n a^t$

CFL

A $L = \{a^n b^n c^m \mid m, n \geq 1\}$

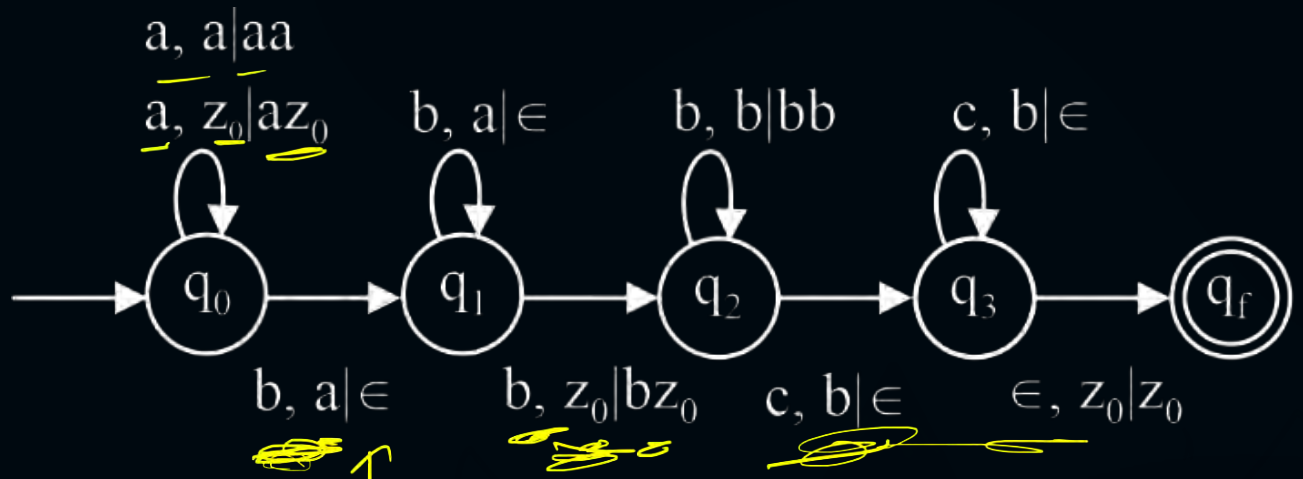
B $L = \{a^n b^n c^m \mid m \leq n\}$ ✗

C $L = \{a^n b^m c^n d^m \mid m, n \geq 0\}$ CSL

D $L = \{a^m b^n c^n d^m \mid m, n \geq 0\}$ CFL

[MCQ]

#Q. The language derived by given PDA is:



A $L = \{a^n b^m c^k \mid n = m + k, m, n, k \geq 0\}$.

B $L = \{a^m b^n c^k \mid k = m + n, m, n, k \geq 1\}$.

C $L = \{a^m b^n c^k \mid k = m + n, m, n, k \geq 0\}$.

D $L = \{a^m b^n c^k \mid n = m + k, m, n, k \geq 1\}$.

[MCQ]

#Q. Suppose L_1 is a finite language and L_2 is non-regular language then $L_1 \cap L_2$ will be:

L_1 = finite → Reg

L_2 = Non-Reg

finite ∩ Non-Reg

Inf.

A. Regular but infinite ✗

B. Non-regular ✗

C. Finite and regular ✓

D. None of these

[MCQ]

#Q. Consider following statements:

S_1: Kleene Closure of infinite set always finite.

S_2: Kleene Closure of finite set always infinite.

Which of the following is correct?

A. S_1 only

B. S_2 only

C. Both S_1 and S_2

D. None of these

[MCQ]

#Q. Consider a language L, then subset of L will be?

$(a+b)^* \Rightarrow \Sigma^*$

$\{a^n b^n \mid n \geq 0\}$

A Regular

B Regular but finite

C Non-Regular

D None of these

[MCQ]

#Q. Let L_1 be regular language and L_2 be CFL language, then which of the following is correct?

A

$L_1 = Reg$
$L_2 = CFL$

Reg ∩ CFL
∪

A. $L_1 \cap L_2$ must be CFL

B. $L_1 - L_2$ must be regular

$L_1 - L_2 \Rightarrow L_1 - L_2$
$\Rightarrow L_1 \cap \overline{L_2}$
Reg ∩ $\overline{CFL}$
R ∩ CSL = CSL

C. $L_1 \subset L_2$ must be regular

D. None of these

[MCQ]

#Q. Consider two languages L_1 and L_2.

If $L_1 \subset L_2$ is regular, then which of the following is true?

A. L_1 must be finite

B. L_1 must be regular

C. L_1 must be non-regular

D. None of these

[MCQ]

#Q. Regular language does not close under on which operation?

(C)

A Complement

B Union

C Subset

D Intersection

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